

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.IT Semester V) MERN

Practical II

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Class: TYIT	Batch:1
Date of Assignment: 30-06-2026	Date/Time of Submission: 30-06-2026

2. Core JavaScript Concepts for MERN

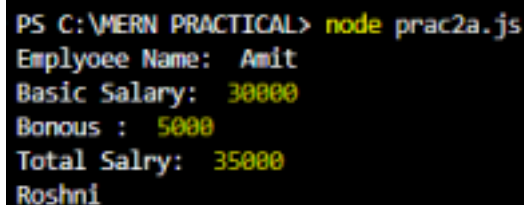
a. Create functions and call them with parameters.

CODE:

```
function calculateSalary name    basicSalary
let bonous = 5000
let totalSalary = basicSalary + bonous
console log "Emplyoee Name: " name
console log "Basic Salary: "    basicSalary
console log "Bonous : "        bonous
console log "Total Salry: "    totalSalary
```

```
calculateSalary "Amit" 30000
console log "Roshni"
```

OUTPUT:



```
PS C:\MERN PRACTICAL> node prac2a.js
Employee Name: Amit
Basic Salary: 30000
Bonous : 5000
Total Salry: 35000
Roshni
```

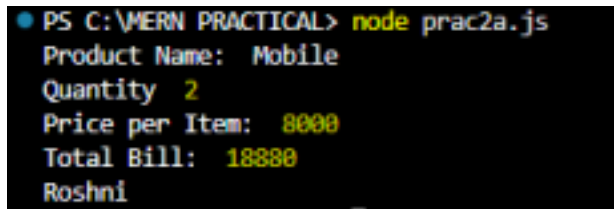
CODE:

```
function genbill product  quantity  price
let total = quantity * price
let totalbill = total + total*0.18
console log "Product Name: " product
console log "Quantity "      quantity
```

```
console log "Price per Item: " price
console log "Total Bill: " totalbill

genbill "Mobile" 2 8000
console log "Roshni"
```

OUTPUT:



```
PS C:\MERN PRACTICAL> node prac2a.js
Product Name: Mobile
Quantity 2
Price per Item: 8000
Total Bill: 18880
Roshni
```

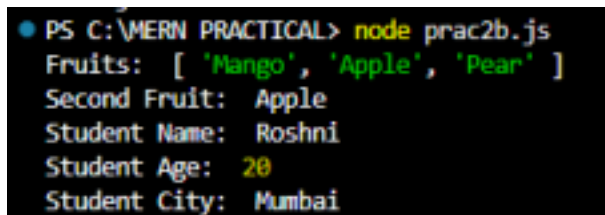
b. Work with arrays and objects in

JavaScript **CODE:**

```
let fruits = "Mango" "Apple" "Pear"
let student =
  name: "Roshni"
  age : 20
  city: "Mumbai"

console log "Fruits: " fruits
console log "Second Fruit: " fruits 1
console log "Student Name: " student name
console log "Student Age: " student age
console log "Student City: " student city
```

OUTPUT:



```
PS C:\MERN PRACTICAL> node prac2b.js
Fruits: [ 'Mango', 'Apple', 'Pear' ]
Second Fruit: Apple
Student Name: Roshni
Student Age: 20
Student City: Mumbai
```

c. Demonstrate use of arrow functions.

CODE:

```
let add = a b =>
  return a + b

console log "SUM = " add 20 10
let sub = a b =>
```

```

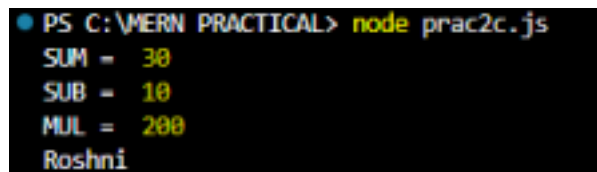
    return a - b

console log "SUB = " sub 20 10
let mul = a b =>
    return a * b

console log "MUL = " mul 20 10
console log "Roshni"

```

OUTPUT:



```

● PS C:\MERN PRACTICAL> node prac2c.js
SUM = 30
SUB = 10
MUL = 200
Roshni

```

d. Write a program using array methods (map, filter, reduce).

CODE:

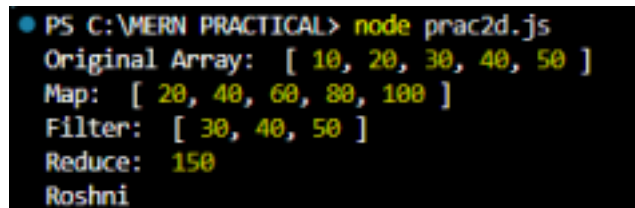
```

let numbers = 10 20 30 40 50
let doubled = numbers map num => num*2    let
greater25 = numbers filter num => num >25    let total
= numbers reduce  sum num  => sum + num 0

console log "Original Array: " numbers
console log "Map: " doubled
console log "Filter: " greater25
console log "Reduce: " total
console log "Roshni"

```

OUTPUT:



```

● PS C:\MERN PRACTICAL> node prac2d.js
Original Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
Roshni

```

e. Create a program that manipulates objects and displays output.

CODE:

```

let student =
  rollno: 41
  name: "Roshni"

```

```

marks: 90

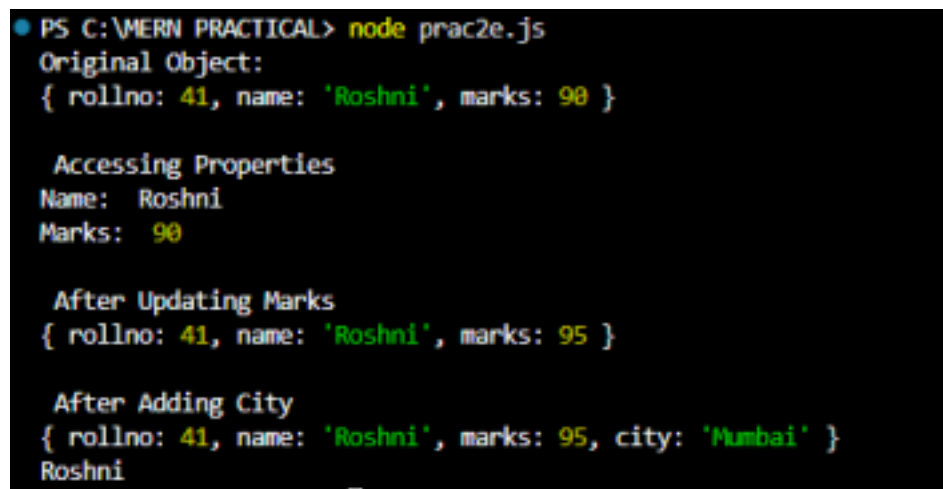
console log "Original Object: "
console log student
console log "\n Accessing Properties"
console log "Name: " student name
console log "Marks: " student marks

student marks = 95
console log "\n After Updating Marks"
console log student

student city = "Mumbai"
console log "\n After Adding City"
console log student
console log "Roshni"

```

OUTPUT:



```

PS C:\MERN PRACTICAL> node prac2e.js
Original Object:
{ rollno: 41, name: 'Roshni', marks: 90 }

Accessing Properties
Name: Roshni
Marks: 90

After Updating Marks
{ rollno: 41, name: 'Roshni', marks: 95 }

After Adding City
{ rollno: 41, name: 'Roshni', marks: 95, city: 'Mumbai' }
Roshni

```

f. Implement a small JavaScript program combining functions, arrays, and objects.

CODE:

```

let students =

rollNo: 101
name: "Rahul"
marks: 85

```

```
rollNo: 102
name: "Varun"
marks: 90
```

```
rollNo: 103
name: "Amit"
marks: 95
```

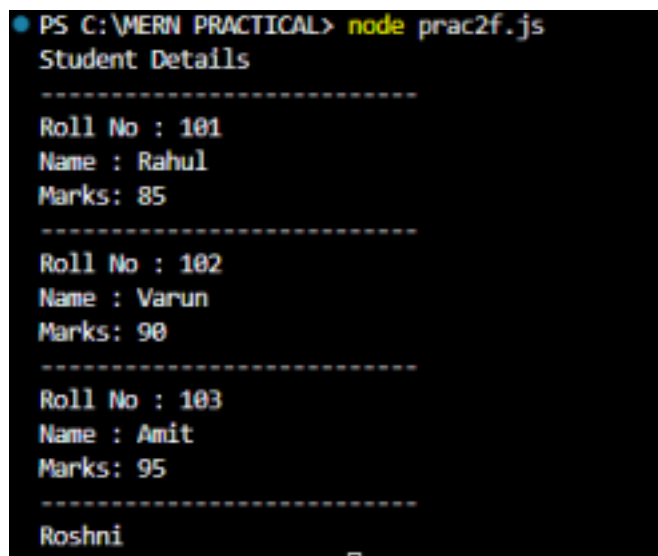
```
function display s

  console log "Student Details"
  console log "-----"
for let stud of s

  console log `Roll No : ${stud rollNo}`
  console log `Name : ${stud name}`
  console log `Marks: ${stud marks}`
  console log "-----"

display students
console log "Roshni"
```

OUTPUT:



```
PS C:\MERN PRACTICAL> node prac2f.js
Student Details
-----
Roll No : 101
Name : Rahul
Marks: 85
-----
Roll No : 102
Name : Varun
Marks: 90
-----
Roll No : 103
Name : Amit
Marks: 95
-----
Roshni
```